

Project Planning Phase

Project Planning (Product Backlog, Sprint Planning, Stories, Story points)

Date	3 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Home Page	USN-1	As a user, I can view the home page to understand the project's purpose and features.	2	High	Kakedu Hareesh
Sprint-1	Prediction	USN-2	As a user, I can enter weather and rainfall details to predict flood risk.	3	High	Kakedu Hareesh
Sprint-1	Prediction	USN-3	As a user, I can submit the input data for flood prediction.	2	High	Kakedu Hareesh
Sprint-1	Prediction Result	USN-4	As a user, I can view whether the prediction is Flood or No Flood .	2	High	Kakedu Hareesh
Sprint-2	Data Processing	USN-6	As a system, I preprocess user input before sending it to the model.	2	High	Kakedu Hareesh
Sprint-2	Result Page	USN-7	As a user, I can view flood safety recommendations based on the prediction.	2	Medium	Kakedu Hareesh
Sprint-2	Error Handling	USN-8	As a user, I receive validation messages for invalid or missing input values.	2	Medium	Kakedu Hareesh
Sprint-3	User Interface	USN-9	As a user, I can use a responsive and attractive web interface on different devices.	3	Medium	Kakedu Hareesh

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	01 Jul 2026	01 Jul 2026	20	01 Jul 2026
Sprint-2	20	6 Days	02 Jul 2026	02 Jul 2026	20	02 Jul 2026
Sprint-3	20	6 Days	03 Jul 2026	03 Jul 2026	20	03 Jul 2026
Sprint-4	20	6 Days	04 Jul 2026	04 Jul 2026	20	04 Jul 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

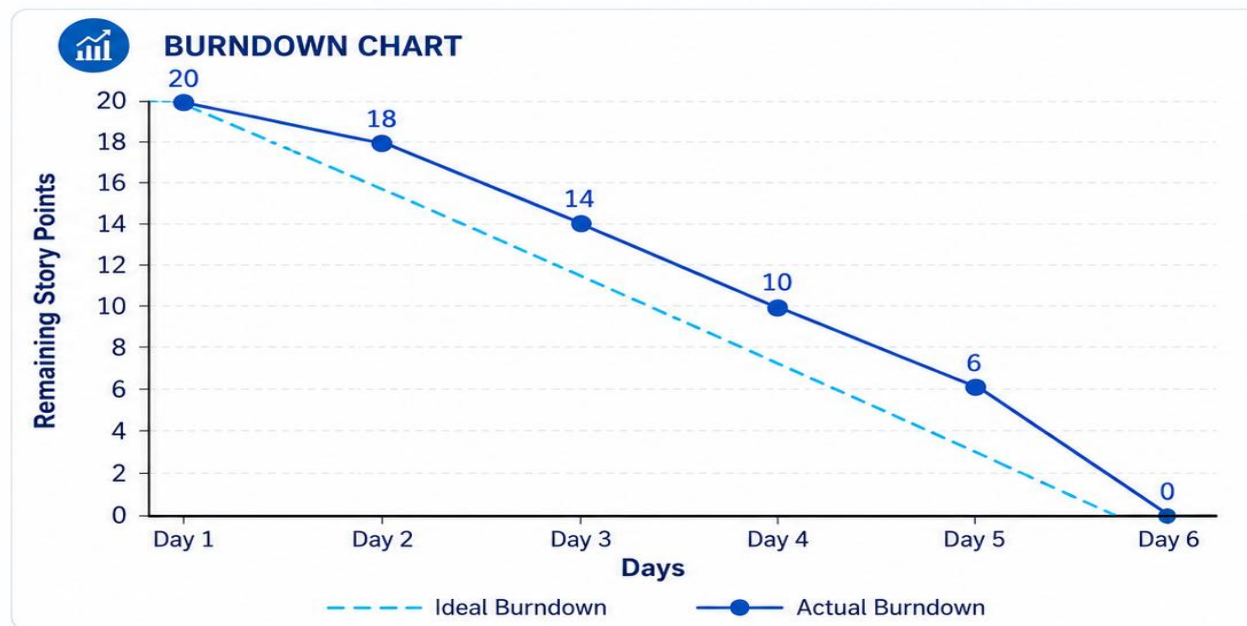
Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.



Sprint Burndown Chart

Rising Waters: Intelligent Flood Prediction and Early Warning System



SPRINT DETAILS	
	Sprint Goal: Develop Flood Prediction Module
	Total Story Points: 20
	Duration: 6 Days
	Model: Decision Tree (96.55% Accuracy)
	WEATHER
	FLOOD
	MACHINE LEARNING
	FLASK

